

Phenomenon #1

for 4 soprano saxophones + electronics

Hunter Brown

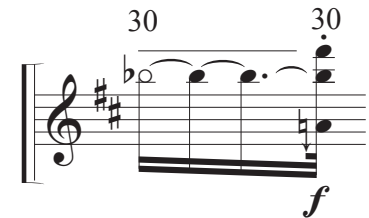
composed in 2020 for ~Nois

Phenomenon #1 uses 4 soprano saxophones as a composite mechanism for directly activating the human hearing apparatus. The saxophone material was recorded and processed through a collection of corrupted audio codecs. The process for corrupting these codecs severely mangles the psychoacoustic algorithms embedded within each codec. When these codecs process audio that contain explicit psychoacoustic phenomena, they produce sounds that challenge universalized notions of auditory perception, while stretching the boundary of what the listener's ear is capable of.

Multiphonics

The multiphonics included in this front matter are indicated in the score like so:

When a multiphonics's number appears over a single note and is followed by a line, the player should use that note as a threshold tone to transition into that multiphonics over the line's duration.



① S/Bb-7



⑫ S/B-5+Eb



⑬ S/Bb-56



⑩ S/Bc-4+Eb



②① S/8G



③⑥ S/B-25



④⑤ S/B-17



②③ S/Eb-3+Bb



③⑦ S/C-25



⑤② S/Bb+EbTc



②⑨ S/B-7+Ta



④② S/Eb-23+Bb



⑤④ S/B-5+Tc



③⑩ S/C#+EbTa



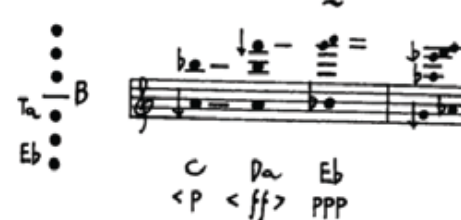
④④ S/Bb-17



⑥⑩ S/B-256+Tc



③① S/Eb+BTa



④⑧ S/Bb-7+Tc



⑦② S/C#-5+c1



Multiphonics cont

66) $\text{S/B-6} + \text{C}_1$

CE Da B
<P <ff> P<f>

76) $\text{S/G} + \text{Bb} + \text{C}_1$

Da Da
mp<f> p<f>

94) $\text{S/C-4} + \text{C}_3$

E Da Da
<mf> <f> <f>

Variable Multiphonics

Variable multiphonics are indicated by square noteheads: smooth transition from a single threshold tone to a multiphonic.

These multiphonics are grouped into 4 categories. Each player should choose 2-3 multiphonics from each category. They should have complete control over whichever multiphonics they choose. They should also alternate between these multiphonics if a number appears back to back in their part. Each player should choose a different set of multiphonics for each category. The categories and some examples of multiphonics in these categories are shown below. Most importantly: the multiphonics should be executed with smooth transitions in and out of threshold tones and with smooth changes in dynamics.

1. Can smoothly transition to and from 1 note to a closely beating cluster. Warm. (pppp-mp):

Examples: 1, 2, 5, 6-8, 10-13, 16, 21, 22, 24, 25, 57, 77, 78, 81, 84, 88, 100, 124, 128, 139

2. Can smoothly transition to and from 1 note to a multiphonic with subtle, microtonal overtones. (pppp-mp):

Examples: 9, 15, 27, 29, 30, 34, 36, 44, 45, 52, 87, 90, 102, 109, 121, 130

3. Can smoothly transition to and from 1 note to a multiphonic with slightly harsher overtones. Like a light distortion effect. Should have a little bite. A bit brighter than 2. (pppp-mp):

Examples: 31, 37, 42, 60, 66, 75, 79, 89, 99

4. Can smoothly transition from 1 note to a consonant cluster (pppp-mp):

Examples: 17-19, 48, 50, 72, 92, 94, 129, 134-138

Sound All notes without a specified accent and/or articulation marking should have a rounded, dull accent followed by a steadily sustained sound. Similar. The "tails" or sustain of each note creates a texture like an undulating ocean of oil and the protruding accents are like bubbles popping out of the murkiness.

Tuning

Sax 1: Sax 2: Sax 3: Sax 4: I want the music to be out of tune.

+43 cents +20 cents -19 cents -35 cents

Don't worry if the detuning messes with your ability to accurately perform multiphonics.

Notation



Trill. If the note is preceded by a grace note, trill those two notes. If there is a trill marking paired with a dyad, trill those two notes

Glissando. Always transform the pitch in a linear shape.

Accidentals carry throughout the measure.



Air leak. Treat this note as a transformation from pitch to air for the duration of the note. Like noise emerging from a pitch. Spit sound is encouraged!

Like so: **Pitch** → **Air/Noise**



Square grace noteheads. Play the grace note on the beat in which the primary note occurs.

Dynamics: Dynamics are relative. Contrast and smooth transitions are the most important. The consistent **pp** is the base level. All sonic transformations should sound like they are emerging from these underlying undulating texture.

Grace note proximity:



Green: Far in front of the primary note. A "wide flam." Should be irregular/trashy/sloppy. **Black/Normal:** Standard treatment of grace note placement



Pink: As close to the primary note as possible. Should sound like a glitch, like you're correcting a wrong note.



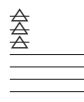
Red Circle note heads. Play these notes 10-20 milliseconds later than written. A reverse tied grace note. Like a glitch in the grid/matrix.



Open slap tongue



Open slap tongue



Teeth on reed: Low to High. Squeal/speaks/sharp shards, always grating!

Electronics Performance Instructions:

Contact me for a Reaper file that contains a click track and the stereo fixed media file: hunterbrownmusic1@gmail.com

Required equipment:

- Audio interface with 4 outputs (outputs 1-2 are routed to the mixer and 3-4 are routed to a headphone amp)
 - Headphone amp with 4 outputs and 4 headphones connected
 - 4 condenser microphones to amplify each saxophone
 - The fixed media output from the interface and the 4 microphones should be routed to a mixer so that an equal balance between the electronics and acoustic saxophones can be achieved on the stereo PA.
 - The stereo PA should be positioned as close to the saxophones as possible. It should sound like the electronics are emanating from the same physical space as the saxophones.
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Stage Layout



Phenomenon #1

for 4 soprano saxophones & electronics

Score

[illegible]

[illegible]

Phenomenon #1

12

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

14

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

p *pp* *mp* *pppp* *mf* *pppp* *pp*

mf *pp* *pppp* *mp* *pp*

pp *p* *pp*

Phenomenon #1

Phenomenon #1

16 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

mf *pp* *mp*

18 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

pp *pp*

Phenomenon #1

20 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

22

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

pppp

pp

mp

pp

pppp

mf

pp

pppp

p

pp

Phenomenon #1

24 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

2

pppp

26 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

p *mp* *pp* *mf* *pp*

Detailed description: The image shows a musical score for a piece titled 'Phenomenon #1'. It consists of two systems of four staves each, labeled S. Sx. 1 through S. Sx. 4. The first system starts at measure 24, and the second system starts at measure 26. The staves are written in treble clef. The first staff (S. Sx. 1) has a key signature of one flat (Bb) and a common time signature. The second staff (S. Sx. 2) has a key signature of one sharp (F#) and a common time signature. The third staff (S. Sx. 3) has a key signature of one flat (Bb) and a common time signature. The fourth staff (S. Sx. 4) has a key signature of one sharp (F#) and a common time signature. The score includes various musical notations such as notes, rests, slurs, and articulation marks. Dynamic markings include *pppp*, *p*, *mp*, *pp*, *mf*, and *pp*. There are also trills and accents marked with 'x' and 'v'. The staves are numbered 24, 26, and 28. The staves are labeled S. Sx. 1, S. Sx. 2, S. Sx. 3, and S. Sx. 4. The staves are written in treble clef. The first staff (S. Sx. 1) has a key signature of one flat (Bb) and a common time signature. The second staff (S. Sx. 2) has a key signature of one sharp (F#) and a common time signature. The third staff (S. Sx. 3) has a key signature of one flat (Bb) and a common time signature. The fourth staff (S. Sx. 4) has a key signature of one sharp (F#) and a common time signature. The score includes various musical notations such as notes, rests, slurs, and articulation marks. Dynamic markings include *pppp*, *p*, *mp*, *pp*, *mf*, and *pp*. There are also trills and accents marked with 'x' and 'v'. The staves are numbered 24, 26, and 28. The staves are labeled S. Sx. 1, S. Sx. 2, S. Sx. 3, and S. Sx. 4. The staves are written in treble clef. The first staff (S. Sx. 1) has a key signature of one flat (Bb) and a common time signature. The second staff (S. Sx. 2) has a key signature of one sharp (F#) and a common time signature. The third staff (S. Sx. 3) has a key signature of one flat (Bb) and a common time signature. The fourth staff (S. Sx. 4) has a key signature of one sharp (F#) and a common time signature. The score includes various musical notations such as notes, rests, slurs, and articulation marks. Dynamic markings include *pppp*, *p*, *mp*, *pp*, *mf*, and *pp*. There are also trills and accents marked with 'x' and 'v'.

[illegible]

Phenomenon #1

32 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

34 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

pp mf p mf pppp

mf pp

mp mf

mf pp

pp f p

pp

Phenomenon #1

41 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

44 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

pppp

pp

pp

pp

12

13

18

pp

p

Detailed description: This musical score is for a piece titled 'Phenomenon #1'. It consists of two systems of four staves each, labeled S. Sx. 1 through S. Sx. 4. The first system covers measures 41 to 43, and the second system covers measures 44 to 46. The music is written in treble clef with a key signature of one sharp (F#). The first system includes various rhythmic patterns, including eighth and sixteenth notes, and rests. The second system introduces a change in time signature from 4/4 to 3/8 in measure 44, which then returns to 4/4. Dynamic markings include 'pppp' (pianissimo) at the start of measure 44, 'pp' (pianissimo) at the start of measure 45, and 'p' (piano) at the start of measure 46. There are also crescendo and decrescendo hairpins. Measure numbers 12, 13, and 18 are indicated above the staves. The score is marked with '8va' (octave up) for the first staff in each system.

48

S. Sx. 1

12

mp *pp*

8^{va}

S. Sx. 2

13

13

(8^{va})

7

3

S. Sx. 3

18

18 (8^{va})

3

5

(8^{va})

S. Sx. 4

mp *pp*

3

3

50

S. Sx. 1

(8^{va})

7

(8^{va})

S. Sx. 2

sffz *pp*

5

3

7

S. Sx. 3

(8^{va})

7

3

5

7

S. Sx. 4

(8^{va})

5

7

Phenomenon #1

52

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

f

pp

fp

pp

8va

8va

8va

8va

3

7

5

3

7

3

54

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

mf

fz pp

pp

mf

8va

8va

8va

8va

3

3

7

5

3

7

3

56 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

pp

ff *pp*

58 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

61 (8va)

S. Sx. 1

5 7 3

72 8va

pp

7

S. Sx. 2

(8va)

47 47 47

pp p pp

5

S. Sx. 3

(8va)

45 45 45

pp p pp

5

S. Sx. 4

(8va)

66 66 66

pp p

64 (8va)

S. Sx. 1

7 3 5 7

7 3 5

S. Sx. 2

(8va)

45 45

pppp

8

S. Sx. 3

(8va)

7 7 7

S. Sx. 4

66 (8va)

3 7 5

pp

Detailed description: This musical score is for a piece titled 'Phenomenon #1' in common time (C). It consists of two systems of four staves each, labeled S. Sx. 1 through S. Sx. 4. The first system covers measures 61 to 72. Measure 61 starts with a treble clef and a key signature of one sharp (F#). The first staff (S. Sx. 1) has a melodic line with eighth notes and rests, with fingerings 5, 7, and 3 indicated. The second staff (S. Sx. 2) has a melodic line with a wavy line indicating a tremolo or rapid oscillation, with a finger number 47. The third staff (S. Sx. 3) has a melodic line with a wavy line and a finger number 45. The fourth staff (S. Sx. 4) has a melodic line with a wavy line and a finger number 66. The second system covers measures 64 to 72. Measure 64 starts with a treble clef and a key signature of one sharp (F#). The first staff (S. Sx. 1) has a melodic line with eighth notes and rests, with fingerings 7, 3, 5, and 7 indicated. The second staff (S. Sx. 2) has a melodic line with a wavy line and a finger number 45. The third staff (S. Sx. 3) has a melodic line with a wavy line and a finger number 7. The fourth staff (S. Sx. 4) has a melodic line with a wavy line and a finger number 66. The score includes various dynamics such as *pp* (pianissimo), *p* (piano), and *pppp* (pianississimo). There are also markings for *8va* (octave up) and *8* (octave down). The tempo is marked as *♩ = 72*.

67 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

pp

pppp pp

70 (8va)

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

sffz fp pp

Phenomenon #1

73 *(8va)*

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

p

pp *mf*

4

7

5

7

4

4

7

5

75 *8va*

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

pp

pp

7

7

5

7

5

7

3

7

77

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

80

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

76

121

mf

p

mp

pp

8va

$fz \text{ } \underline{pp}$

sfz pp

$$pp < ff$$

95

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

128

121

102

p *ff* *pp*

pp *p* *pp*

mp *pppp*

99 (8^{va}) 57

S. Sx. 1

7 *ppp* *p* *pp*

S. Sx. 2

(8^{va}) *ppp* *pp*

S. Sx. 3

128 *mp* *pp*

S. Sx. 4

mp

104 D

S. Sx. 1

pp

S. Sx. 2

(8^{va}) 45 *pppp*

S. Sx. 3

52 *pppp*

S. Sx. 4

pppp *pp*

Phenomenon #1

110 50 29 imperceptible attack/ emerge slowly

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

p

pppp

mp

ppp *p* *pppp* *pp*

mf *pp*

117 52 22

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

p

pp *pppp* *p*

pppp *p*

Phenomenon #1

rit.

$\text{♩} = 52$

126

S. Sx. 1

S. Sx. 2

18

pppp

52

p

S. Sx. 3

S. Sx. 4

no breath/break

E

136

S. Sx. 1

p

S. Sx. 2

p

pppp

S. Sx. 3

p

S. Sx. 4

p

143

S. Sx. 1



imperceptible attack/
45 emerge slowly

S. Sx. 2



pppp

S. Sx. 3



S. Sx. 4



151 (Repeat 3x)

S. Sx. 1

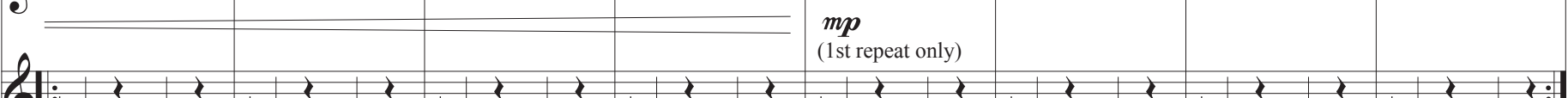


S. Sx. 2



mp
(1st repeat only)

S. Sx. 3



S. Sx. 4



159

S. Sx. 1

S. Sx. 2

S. Sx. 3

S. Sx. 4

This musical score is for a four-part setting of a text, labeled 'Phenomenon #1'. It begins at measure 159. The parts are Soprano Saxophone 1 (S. Sx. 1), Soprano Saxophone 2 (S. Sx. 2), Alto Saxophone 3 (S. Sx. 3), and Alto Saxophone 4 (S. Sx. 4). The key signature has one flat (B-flat), and the time signature is 4/4. In measure 159, S. Sx. 1 has a half note G4 with a bar line above it, followed by a quarter rest. S. Sx. 2 has a whole note chord of G4 and F#4, with a large slur over it. S. Sx. 3 has a half note F#3 with a bar line below it, followed by a quarter rest. S. Sx. 4 has a half note F#3 with a bar line below it, followed by a quarter rest. In measure 160, S. Sx. 1 has a half note G4 with a bar line above it, followed by a quarter rest. S. Sx. 2 has a whole note chord of G4 and F#4, with a large slur over it. S. Sx. 3 has a half note F#3 with a bar line below it, followed by a quarter rest. S. Sx. 4 has a half note F#3 with a bar line below it, followed by a quarter rest. In measure 161, S. Sx. 1 has a half note G4 with a bar line above it, followed by a quarter rest. S. Sx. 2 has a whole note chord of G4 and F#4, with a large slur over it. S. Sx. 3 has a half note F#3 with a bar line below it, followed by a quarter rest. S. Sx. 4 has a half note F#3 with a bar line below it, followed by a quarter rest. In measure 162, S. Sx. 1 has a half note G4 with a bar line above it, followed by a quarter rest. S. Sx. 2 has a whole note chord of G4 and F#4, with a large slur over it. S. Sx. 3 has a half note F#3 with a bar line below it, followed by a quarter rest. S. Sx. 4 has a half note F#3 with a bar line below it, followed by a quarter rest. The score ends with a double bar line at the end of measure 162.